

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Balance (BAX_L)

Data File: BAX_L.xpt

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Component Description

The ability to maintain balance is essential to nearly all activities associated with daily living. The balance system enables us to sense where we are in space and to maintain our posture and equilibrium while we are still or moving. Impairment of the balance system can be a major disabling condition for those who are affected. The control of balance requires the integration of information from multiple sensory and motor systems by the central nervous system (CNS). Balance receptors in the inner ear (the vestibular system) provide information to the CNS about head and body movements. The eyes (visual system) provide input regarding the body's orientation within the environment and about motion within the environment. The position and motion sensors of the muscles and joints, and the touch receptors of the extremities (proprioceptive system), send signals regarding bodily position, particularly in relation to the support surface. The CNS integrates all this information, determines the body's spatial orientation, and sends appropriate neural messages to the motor system to activate movements that will maintain equilibrium and keep our vision clear when we are moving.

In NHANES August 2021–August 2023, the modified Romberg Test of Standing Balance on Firm and Complaint Support Surfaces (MRT) (Weber, 1993) was included.

Eligible Sample

Participants aged 20 to 69 years were eligible. Participants were excluded if they: 1) had vision so poor that they needed assistance to find their way to the test room; 2) were pregnant; 3) had an amputation of the leg or feet (other than toes); 4) weighed more than 315 pounds, since the foam used to stand on during the test may become too compressed to provide a suitable surface; 5) were unable to stand on their own; 6) had a prosthetic device below the waist; 7) stated that they are unable to do the MRT because of a recent injury or surgery to the legs, ankles, or feet; 8) were experiencing both current symptoms of dizziness, lightheadedness or feeling faint and had fallen in the past 12 months because of problems with dizziness or balance; 9) were wearing shoes with heels 3 inches or more; 10) had a body size either too small or too large (determined when the safety belt was put on) to accommodate proper fitting of the safety belt; or 11) said that they were not okay to begin the balance test and either wished to speak to a physician (physician referral) or did not wish to speak to a physician. In addition, participants were excluded only from Condition 5 of the MRT if they: 1) had neck pain at time of exam; 2) ever had neck surgery; 3) ever had a neck problem that lasted more than 6 weeks; or 4) could not move their head and neck in the range of motion required for the condition.

Protocol and Procedure

The MRT included a maximum of five test conditions of increasing difficulty, and participants were permitted up to two trials for each condition. Because each successive condition was progressively more difficult than the condition preceding it, the MRT testing ended whenever a subject failed to pass a test condition after two attempted trials. Participants were asked to wear a gait belt with handles (i.e., a device worn to help people with mobility problems) around their waists so the technician could help stabilize them, if needed. While most

participants were in MEC-provided slip-resistant socks, if wearing shoes, the participants continued if the heel was less than 3 inches.

In all MRT conditions, the participants were asked to stand with feet together (with allowances made as needed for orthopedic conditions) and arms crossed at the chest. In Condition 1, the participant stood on the bare floor with eyes open. Condition 2 involved standing on the bare floor with eyes closed. Condition 3 involved standing on a dense foam surface with eyes open. Condition 4 involved standing on a dense foam surface with eyes closed. Condition 5 involved standing on a dense foam surface with eyes closed and moving the head slowly side-to-side.

For all conditions, the participant passed the condition if they were able to maintain balance for a requisite period of time. The technician monitored the SP for failure to maintain balance, such as movement of the feet, significant movement of the arms, opening the eyes (for Conditions 2, 4, or 5), or a need for technician intervention due to excessive torso motion (e.g., sway) while attempting to maintain balance. A failure for the condition was documented if the SP failed to maintain balance for the requisite time interval.

The requisite time intervals differed across conditions. The time interval for Conditions 1-2 was 15 seconds; and 15 seconds was considered a passing score. The time interval for Conditions 3-5 was 30 seconds, but 20 seconds was considered a passing score. For Conditions 3-5, participants who passed Trial 1 with a score of 20-29 seconds were offered Trail 2 of the condition. For Conditions 3-5, the time interval was 30 seconds to match previous data collection (NHANES 1999-2004). However, to match current research on falls risk assessment, a "pass" was established if the participant was able to maintain balance for 20 seconds (2019, Cohen).

Did not pass (DNP) was defined as any occurrence of any of the following before 15 seconds for Conditions 1 and 2, or 20 seconds for Conditions 3-5: 1) movement of the participant's feet from the initial test position (falling, side stepping, hopping, pivoting, etc.); 2) movement of the participant's arms off of the chest (or waist); 3) participant's eyes open during an eyes-closed condition (i.e., Conditions 2, 4, and 5); 4) touching the wall; or 5) any intervention by the examiner to stop the participant from falling.

Additional details on the protocol are described in the [NHANES August 2021–August 2023 procedure manuals](#).

Quality Assurance & Quality Control

Technicians were trained in the testing protocols and technicians were periodically observed to verify that standard balance testing procedures were being strictly followed. All data were directly entered by NHANES technicians into ISIS, a computerized database information system. Quality control procedures were performed to check for completeness and data validity and to identify logical inconsistencies. Health technician performance was also monitored using data reviews.

Data Processing and Editing

Prior to data release, NHANES staff reviewed data for completeness and logical inconsistencies and cross-checked with other available data for verification and reviewed free-text comments noted by MEC staff and made edits or corrections as appropriate.

The balance exam status code (BAXMSTAT) was created and indicates the following for each participant:

- 1 = Complete exam (i.e., completed the maximum duration for all eligible conditions or attempted both trials of all eligible conditions).
- 2 = Partial exam (i.e., started the exam but did not complete because of refusal; limited time during exam visit; physical limitation or ill/emergency; equipment failure or error; or other).
- 3 = Not done (i.e., did not start the exam because of refusal; limited time during exam visit;

physical limitation or ill/emergency; equipment failure or error; or other).
4 = Ineligible for all MRT conditions (see eligibility criteria above).

The BAXMSTAT variable identifies all participants who are ineligible for the balance exam (BAXMSTAT=4). Additionally, some variables for common reasons for ineligibility to the balance exam are included in this file (i.e., BAQ110, BAQ125, BAQ132, BAQ140, BAQ150, BAQ160, BAQ170), but other variables for ineligibility are not separately included on the datafile primarily due to potential disclosure concerns. Although exclusion criteria for Conditions 1-4 applied to Condition 5, there were additional exclusion criteria for Condition 5 and a status code for Condition 5 (BAX5STAT) was created. Reason codes for partial exams (BAXRXNC) and exams not done (BAXRXND) were created from MEC staff comments and included in this data release.

Analytic Notes

NHANES conducted an MRT in 1999-2004, but there are differences between the 1999-2004 and the protocol implemented during 2019-March 2020 and August 2021–August 2023. Differences include the following: targeted age range; exclusion criteria; Conditions 3 and Condition 4 required 30 seconds, rather than 20 seconds to pass in 1999-2004; and Condition 5 was not assessed in 1999-2004. In 1999-2004, the dataset included a failure time for participants who “did not pass” a condition, but in 2019-March 2020 and August 2021–August 2023, the dataset includes duration time for participants who attempted a condition. Thus, if an analyst wanted to compare more directly the 1999-2004 and August 2021–August 2023 data files, they may consider recoding the participants in August 2021–August 2023 who passed Condition 3 or Condition 4 with a duration of 20-29 seconds as “did not pass”. To alert data users to differences between 1999-2004 and August 2021–August 2023, the variables corresponding to “pass”/“did not pass” have a modified naming convention (i.e., BAXPFC11 in 1999-2004 and BAXPF11 in August 2021–August 2023 for Condition 1, Trial 1). Similarly, the variables corresponding to time or duration have a modified naming convention (i.e., BAXFTC11 in 1999-2004 and BAXTC11 in August 2021–August 2023 for Condition 1, Trial 1).

Exam sample weights should be used for analyses. Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

References

- Cohen HS, Mulavara AP, Stitz J, Sangi-Haghpeykar H, Williams SP, Peters BT, Bloomberg JJ (2019). Screening for vestibular disorders using the modified clinical test of sensory interaction and balance and tandem walking with eyes closed. *Otol Neurotol*, 40, 658-665. DOI: 10.1097/MAO.0000000000002173.
- Weber PC, Cass SP. Clinical assessment of postural stability. *Am J Otol*. 1993;14:566-9.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 20 YEARS - 69 YEARS

BAXMSTAT - MRT exam status

Variable Name: BAXMSTAT

SAS Label: MRT exam status

English Text: MRT exam status

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Complete	3732	3732	BAX5STAT
2	Partial	219	3951	BAXRXNC
3	Not done	213	4164	BAXRXND
4	Ineligible for all MRT conditions	607	4771	BAX5STAT
.	Missing	0	4771	

BAXRXNC - Reason for partial MRT exam

Variable Name: BAXRXNC

SAS Label: Reason for partial MRT exam

English Text: Reason for partial MRT exam

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP refusal	176	176	BAX5STAT
2	Limited time	1	177	BAX5STAT
3	Physical limitation or SP ill/emergency	5	182	BAX5STAT
4	Equipment failure or error (technician/software/supply)	17	199	BAX5STAT
5	Other	5	204	BAX5STAT
.	Missing	4567	4771	

BAXRXND - Reason for not done MRT exam**Variable Name:** BAXRXND**SAS Label:** Reason for not done MRT exam**English Text:** Reason for not done MRT exam**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP refusal	9	9	BAX5STAT
2	Limited time	103	112	BAX5STAT
3	Physical limitation or SP ill/emergency	26	138	BAX5STAT
4	Equipment failure or error (technician/software/supply)	5	143	BAX5STAT
5	Other	70	213	BAX5STAT
.	Missing	4558	4771	

BAX5STAT - Eligibility for Condition 5

Variable Name: BAX5STAT
SAS Label: Eligibility for Condition 5
English Text: Eligibility for Condition 5
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Eligible for Condition 5	3248	3248	
2	Ineligible for Condition 5	1317	4565	
.	Missing	206	4771	

BAQ110 - Can stand on your own?**Variable Name:** BAQ110**SAS Label:** Can stand on your own?**English Text:** Can stand on your own?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	4305	4305	
2	No	58	4363	
7	Refused	0	4363	
9	Don't know	0	4363	
.	Missing	408	4771	

BAQ121 - Are you currently wearing a leg brace?**Variable Name:** BAQ121**SAS Label:** Are you currently wearing a leg brace?**English Text:** Are you currently wearing a leg brace?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	25	25	
2	No	4279	4304	
7	Refused	1	4305	
9	Don't know	0	4305	
.	Missing	466	4771	

BAQ125 - Injury to foot, leg, or hip

Variable Name: BAQ125

SAS Label: Injury to foot, leg, or hip

English Text: Injury to foot, leg, or hip

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	253	253	
2	No	4024	4277	
7	Refused	0	4277	
9	Don't know	2	4279	
.	Missing	492	4771	

BAQ132 - Problems with dizzy, past 24 hours

Variable Name: BAQ132

SAS Label: Problems with dizzy, past 24 hours

English Text: Problems with dizzy, past 24 hours

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	200	200	
2	No	3822	4022	
7	Refused	1	4023	
9	Don't know	1	4024	
.	Missing	747	4771	

BAQ140 - Last 12 months, fallen due to dizziness**Variable Name:** BAQ140**SAS Label:** Last 12 months, fallen due to dizziness**English Text:** Last 12 months, fallen due to dizziness**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	261	261	
2	No	3760	4021	
7	Refused	1	4022	
9	Don't know	2	4024	
.	Missing	747	4771	

BAQ150 - Do you have neck pain now?**Variable Name:** BAQ150**SAS Label:** Do you have neck pain now?**English Text:** Do you have neck pain now?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	354	354	
2	No	3618	3972	
7	Refused	0	3972	
9	Don't know	0	3972	
.	Missing	799	4771	

BAQ160 - Have you ever had surgery on your neck?**Variable Name:** BAQ160**SAS Label:** Have you ever had surgery on your neck?**English Text:** Have you ever had surgery on your neck?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	103	103	
2	No	3515	3618	
7	Refused	0	3618	
9	Don't know	0	3618	
.	Missing	1153	4771	

BAQ170 - Neck problem lasted more than 6 weeks**Variable Name:** BAQ170**SAS Label:** Neck problem lasted more than 6 weeks**English Text:** Neck problem lasted more than 6 weeks**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	228	228	
2	No	3287	3515	
7	Refused	0	3515	
9	Don't know	0	3515	
.	Missing	1256	4771	

BAQ201 - Can you move head 30 degrees left/right?**Variable Name:** BAQ201**SAS Label:** Can you move head 30 degrees left/right?**English Text:** Can you move head 30 degrees left/right?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	3259	3259	
2	No	28	3287	
7	Refused	0	3287	
9	Don't know	0	3287	
.	Missing	1484	4771	

BAQ173 - Are you okay to begin this balance test?**Variable Name:** BAQ173**SAS Label:** Are you okay to begin this balance test?**English Text:** Are you okay to begin this balance test?**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	3964	3964	
2	No	4	3968	
7	Refused	0	3968	
9	Don't know	0	3968	
.	Missing	803	4771	

BAXPF11 - MRT cond. 1, trial 1, eyes open**Variable Name:** BAXPF11**SAS Label:** MRT cond. 1, trial 1, eyes open**English Text:** MRT cond. 1, trial 1, eyes open**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	3902	3902	
2	DNP (did not pass)	49	3951	
7	Refused	0	3951	
.	Missing	820	4771	

BAXTC11 - MRT cond. 1, trial 1, time

Variable Name: BAXTC11
SAS Label: MRT cond. 1, trial 1, time
English Text: MRT cond. 1, trial 1, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15	Range of Values	3950	3950	
.	Missing	821	4771	

BAARFC11 - MRT cond. 1, trial 1, reason stopped

Variable Name: BAARFC11
SAS Label: MRT cond. 1, trial 1, reason stopped
English Text: MRT cond. 1, trial 1, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	14	14	
2	Arms moved off waist	4	18	
3	Eyes opened	0	18	
4	Tech intervention	28	46	
5	Grab/touch wall	3	49	
.	Missing	4722	4771	

BAXPF12 - MRT cond. 1, trial 2, eyes open**Variable Name:** BAXPF12**SAS Label:** MRT cond. 1, trial 2, eyes open**English Text:** MRT cond. 1, trial 2, eyes open**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	43	43	
2	DNP (did not pass)	5	48	
7	Refused	0	48	
.	Missing	4723	4771	

BAXTC12 - MRT cond. 1, trial 2, time

Variable Name: BAXTC12
SAS Label: MRT cond. 1, trial 2, time
English Text: MRT cond. 1, trial 2, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2 to 15	Range of Values	48	48	
.	Missing	4723	4771	

BAARFC12 - MRT cond. 1, trial 2, reason stopped

Variable Name: BAARFC12
SAS Label: MRT cond. 1, trial 2, reason stopped
English Text: MRT cond. 1, trial 2, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	1	1	
2	Arms moved off waist	1	2	
3	Eyes opened	0	2	
4	Tech intervention	2	4	
5	Grab/touch wall	1	5	
.	Missing	4766	4771	

BAXPF21 - MRT cond. 2, trial 1, eyes closed**Variable Name:** BAXPF21**SAS Label:** MRT cond. 2, trial 1, eyes closed**English Text:** MRT cond. 2, trial 1, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	3846	3846	
2	DNP (did not pass)	96	3942	
7	Refused	1	3943	
.	Missing	828	4771	

BAXTC21 - MRT cond. 2, trial 1, time

Variable Name: BAXTC21

SAS Label: MRT cond. 2, trial 1, time

English Text: MRT cond. 2, trial 1, time

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15	Range of Values	3942	3942	
.	Missing	829	4771	

BAARFC21 - MRT cond. 2, trial 1, reason stopped

Variable Name: BAARFC21
SAS Label: MRT cond. 2, trial 1, reason stopped
English Text: MRT cond. 2, trial 1, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	29	29	
2	Arms moved off waist	4	33	
3	Eyes opened	42	75	
4	Tech intervention	9	84	
5	Grab/touch wall	10	94	
.	Missing	4677	4771	

BAXPF22 - MRT cond. 2, trial 2, eyes closed**Variable Name:** BAXPF22**SAS Label:** MRT cond. 2, trial 2, eyes closed**English Text:** MRT cond. 2, trial 2, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	69	69	
2	DNP (did not pass)	26	95	
7	Refused	2	97	
.	Missing	4674	4771	

BAXTC22 - MRT cond. 2, trial 2, time

Variable Name: BAXTC22
SAS Label: MRT cond. 2, trial 2, time
English Text: MRT cond. 2, trial 2, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15	Range of Values	94	94	
.	Missing	4677	4771	

BAARFC22 - MRT cond. 2, trial 2, reason stopped

Variable Name: BAARFC22
SAS Label: MRT cond. 2, trial 2, reason stopped
English Text: MRT cond. 2, trial 2, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	11	11	
2	Arms moved off waist	3	14	
3	Eyes opened	7	21	
4	Tech intervention	1	22	
5	Grab/touch wall	4	26	
.	Missing	4745	4771	

BAXPF31 - MRT cond. 3, trial 1, eyes open**Variable Name:** BAXPF31**SAS Label:** MRT cond. 3, trial 1, eyes open**English Text:** MRT cond. 3, trial 1, eyes open**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	3820	3820	
2	DNP (did not pass)	66	3886	
7	Refused	11	3897	
.	Missing	874	4771	

BAXTC31 - MRT cond. 3, trial 1, time

Variable Name: BAXTC31

SAS Label: MRT cond. 3, trial 1, time

English Text: MRT cond. 3, trial 1, time

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	3882	3882	
.	Missing	889	4771	

BAARFC31 - MRT cond. 3, trial 1, reason stopped

Variable Name: BAARFC31
SAS Label: MRT cond. 3, trial 1, reason stopped
English Text: MRT cond. 3, trial 1, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	24	24	
2	Arms moved off waist	13	37	
3	Eyes opened	0	37	
4	Tech intervention	21	58	
5	Grab/touch wall	22	80	
.	Missing	4691	4771	

BAXPF32 - MRT cond. 3, trial 2, eyes open**Variable Name:** BAXPF32**SAS Label:** MRT cond. 3, trial 2, eyes open**English Text:** MRT cond. 3, trial 2, eyes open**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	63	63	
2	DNP (did not pass)	16	79	
7	Refused	1	80	
.	Missing	4691	4771	

BAXTC32 - MRT cond. 3, trial 2, time

Variable Name: BAXTC32
SAS Label: MRT cond. 3, trial 2, time
English Text: MRT cond. 3, trial 2, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	79	79	
.	Missing	4692	4771	

BAARFC32 - MRT cond. 3, trial 2, reason stopped

Variable Name: BAARFC32
SAS Label: MRT cond. 3, trial 2, reason stopped
English Text: MRT cond. 3, trial 2, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	8	8	
2	Arms moved off waist	1	9	
3	Eyes opened	0	9	
4	Tech intervention	3	12	
5	Grab/touch wall	8	20	
.	Missing	4751	4771	

BAXPF41 - MRT cond. 4, trial 1, eyes closed**Variable Name:** BAXPF41**SAS Label:** MRT cond. 4, trial 1, eyes closed**English Text:** MRT cond. 4, trial 1, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	2591	2591	
2	DNP (did not pass)	1260	3851	
7	Refused	12	3863	
.	Missing	908	4771	

BAXTC41 - MRT cond. 4, trial 1, time**Variable Name:** BAXTC41**SAS Label:** MRT cond. 4, trial 1, time**English Text:** MRT cond. 4, trial 1, time**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30	Range of Values	3850	3850	
.	Missing	921	4771	

BAARFC41 - MRT cond. 4, trial 1, reason stopped

Variable Name: BAARFC41
SAS Label: MRT cond. 4, trial 1, reason stopped
English Text: MRT cond. 4, trial 1, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	366	366	
2	Arms moved off waist	103	469	
3	Eyes opened	372	841	
4	Tech intervention	143	984	
5	Grab/touch wall	406	1390	
.	Missing	3381	4771	

BAXPF42 - MRT cond. 4, trial 2, eyes closed**Variable Name:** BAXPF42**SAS Label:** MRT cond. 4, trial 2, eyes closed**English Text:** MRT cond. 4, trial 2, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	690	690	
2	DNP (did not pass)	631	1321	
7	Refused	65	1386	
.	Missing	3385	4771	

BAXTC42 - MRT cond. 4, trial 2, time

Variable Name: BAXTC42
SAS Label: MRT cond. 4, trial 2, time
English Text: MRT cond. 4, trial 2, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30	Range of Values	1319	1319	
.	Missing	3452	4771	

BAARFC42 - MRT cond. 4, trial 2, reason stopped

Variable Name: BAARFC42
SAS Label: MRT cond. 4, trial 2, reason stopped
English Text: MRT cond. 4, trial 2, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	174	174	
2	Arms moved off waist	55	229	
3	Eyes opened	142	371	
4	Tech intervention	88	459	
5	Grab/touch wall	245	704	
.	Missing	4067	4771	

BAXPF51 - MRT cond. 5, trial 1, eyes closed**Variable Name:** BAXPF51**SAS Label:** MRT cond. 5, trial 1, eyes closed**English Text:** MRT cond. 5, trial 1, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	1501	1501	
2	DNP (did not pass)	1119	2620	
7	Refused	15	2635	
.	Missing	2136	4771	

BAXTC51 - MRT cond. 5, trial 1, time

Variable Name: BAXTC51

SAS Label: MRT cond. 5, trial 1, time

English Text: MRT cond. 5, trial 1, time

Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30	Range of Values	2595	2595	
.	Missing	2176	4771	

BAARFC51 - MRT cond. 5, trial 1, reason stopped

Variable Name: BAARFC51
SAS Label: MRT cond. 5, trial 1, reason stopped
English Text: MRT cond. 5, trial 1, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	444	444	
2	Arms moved off waist	106	550	
3	Eyes opened	184	734	
4	Tech intervention	151	885	
5	Grab/touch wall	402	1287	
.	Missing	3484	4771	

BAXPF52 - MRT cond. 5, trial 2, eyes closed**Variable Name:** BAXPF52**SAS Label:** MRT cond. 5, trial 2, eyes closed**English Text:** MRT cond. 5, trial 2, eyes closed**Target:** Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pass	541	541	
2	DNP (did not pass)	683	1224	
7	Refused	24	1248	
.	Missing	3523	4771	

BAXTC52 - MRT cond. 5, trial 2, time

Variable Name: BAXTC52
SAS Label: MRT cond. 5, trial 2, time
English Text: MRT cond. 5, trial 2, time
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30	Range of Values	1223	1223	
.	Missing	3548	4771	

BAARFC52 - MRT cond. 5, trial 2, reason stopped

Variable Name: BAARFC52
SAS Label: MRT cond. 5, trial 2, reason stopped
English Text: MRT cond. 5, trial 2, reason stopped
Target: Both males and females 20 YEARS - 69 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Feet moved	256	256	
2	Arms moved off waist	55	311	
3	Eyes opened	92	403	
4	Tech intervention	125	528	
5	Grab/touch wall	262	790	
.	Missing	3981	4771	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Body Measures (BMX_L)

Data File: BMX_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

NHANES body measures data are used to monitor trends in infant and child growth, to estimate the prevalence of overweight and obesity in U.S. children, adolescents, and adults, and to examine the associations between body weight and the health and nutritional status of the U.S. population.

The measurements and target age groups for the public body measures data file (BMX_L) are as follows:

- Weight: All ages
- Head circumference: birth through 6 months
- Recumbent length: birth through 47 months
- Standing height: 2 years and older
- Upper leg length: 8 years and older
- Upper arm length: 2 months and older
- Mid-upper arm circumference: 2 months and older
- Waist circumference: 2 years and older
- Hip circumference: 12 years and older

Eligible Sample

All examined survey participants in the NHANES August 2021-August 2023 sample are included in this dataset. There were no medical, safety, or other exclusions for the body measures protocol. The health technicians used their discretion to obtain as many measures as practical for people who used a wheelchair.

Protocol and Procedure

The body measures data were collected, in the Mobile Examination Center (MEC), by trained health technicians. The health technician was assisted by a recorder during the body measures examination. The participant's age at the time of the screening interview determined the body measures examination protocol. In some instances, the age at the screening interview and age at the time of the health examination differed by several weeks. The Demographics data file includes variables age in years at screening (RIDAGEYR) for all participants and age in months at exam (RIDEXAGEM) for participants 0 to 239 months.

Arm and leg measurements were made on the right side of the body. If a participant had an amputation, medical condition, or medical appliance, such as a cast, that prevented measurements from being taken on the right side of the body, the health technician took measurements on the left side. The body measures file does not identify participants who had amputations because that information may be considered identifiable and pose a disclosure

concern. The body weight data for participants who had limb amputations were set to "missing."

This data file includes body measures for women who were pregnant at the time of their health examination. Pregnancy status at the time of the health examination is indicated by the variable, RIDEXPRG, in the Demographic data file. RIDEXPRG values are reported for women 20–44 years of age. RIDEXPRG for several pregnant women who were outside of this age range are not reported due to disclosure concerns. The body measures data for these participants are also not reported.

The [NHANES Anthropometry Procedures Manual](#) are available on the NHANES website.

Quality Assurance & Quality Control

The NHANES health technicians completed a 2-day training program with survey staff and an expert anthropometrist. The training included an overview of the component, general guidelines and technical skills for each measurement, and demonstrations conducted by the expert examiner with volunteer subjects. The expert examiner reviewed and demonstrated the proper technique to use for each measure. Supervised practice exercises followed, conducted with several volunteer subjects, including infants, children, and adults. The chief health technician, at each of the MECs, monitored staff performance in the field. Health technician performance was also monitored using direct observation, data reviews, and periodic expert examiner (gold standard comparison) evaluations.

The body measures examination rooms in each of the MECs were identical with respect to layout and equipment. Scheduled equipment calibration was performed by the health technicians and verified by supervisory staff. The Anthropometry Procedures Manual includes detailed descriptions of the quality assurance and quality control measures that are used in the NHANES anthropometry/body measures component.

Data Processing and Editing

The BMX_L data were reviewed for unusual and erroneous values. Review criteria were based on the NHANES 1999–2016 body measurement data. During the data review, values that were above the 99th percentile or below the 1st percentile, for a specific age or age-gender group, were flagged for review. When records were flagged, the entire body measures record was reviewed for reasonableness. Subject characteristics, such as height, weight, age, and gender were taken into consideration. Values that were determined to be unrealistic were deleted from the file. None of the original body measures data were changed and there are no imputed values in this file.

Component status code (BMXSTATS):

This variable was created to indicate body measures component status. There are four categories:

1. Complete data for age group
2. Partial: only height and weight obtained
3. Other: partial exam
4. No body measures exam data

Body Mass Index (BMXBMI):

Body Mass Index (BMI) was calculated as weight in kilograms divided by height in meters squared, and then rounded to one decimal place.

BMI Category – Children/Adolescents (BMDBMIC):

This variable was created for children and adolescents aged 2 to 19 years at examination. Cutoff criteria are based on the Centers for Disease Control (CDC) growth chart "[BMI-for-age charts, 2 to 20 years, by sex and age](#)". Age in months at examination was used to match age

in months from BMI growth chart data, separately for males and females. There are four categories:

1. Underweight (BMI < 5th percentile)
2. Normal weight (BMI 5th to < 85th percentiles)
3. Overweight (BMI 85th to < 95th percentiles)
4. Obese (BMI ≥ 95th percentile)

Analytic Notes

Component status code: A final body measures component status code provides analysis with a quick method of identifying survey participants with complete or partial body measure measurement data.

Unusual values: Unusual body measures values were noted during the review of the data. Typically, unusual values occurred when a subject was extremely short, tall, overweight, or underweight. In addition, the upper arm length (BMXARML) and upper leg length (BMXLEG) values may be affected by extreme amounts of adipose tissue. Analysts should examine the distributions of the body measurements carefully and consider whether it is appropriate to include or exclude extreme values in each analysis.

Comment codes: Comment codes were added by the health technicians, during data collection, to document problems or situations that arose during the body measures examination. For example, the variable BMIWT is a comment code for the body weight measurement. If a participant did not change into the NHANES exam gown, a code of BMIWT of "3" denoting "clothing worn" was made in the record. Analysts should review the comment code information for each of the body measures prior to data analysis.

Weight status classification: BMI, expressed as weight in kilograms divided by height in meters squared (Kg/m^2), is commonly used to classify weight status. The definitions of underweight, normal weight, overweight, and obesity in children and adolescents are not directly comparable with the definitions in adults. The age- and sex-specific 5th, 85th, and 95th percentiles of the growth charts are usually used as cutoff criteria for children and adolescents. The variable BMDBMIC provides weight status categorical for children and adolescents aged 2 to 19 years at examination. Consequently, BMDBMIC was not calculated for a few participants who were 19 years at the screening interview but became 20 years at the health examination.

Sample weights: The NHANES examination sample weights should be used to analyze the body measures data. Please refer to the [NHANES Analytic Guidelines](#) and the online [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

BMDSTATS - Body Measures Component Status Code

Variable Name: BMDSTATS
SAS Label: Body Measures Component Status Code
English Text: Body Measures Component status Code
Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Complete data for age group	8235	8235	
2	Partial: Only height and weight obtained	147	8382	
3	Other partial exam	409	8791	
4	No body measures exam data	69	8860	
.	Missing	0	8860	

BMXWT - Weight (kg)

Variable Name: BMXWT

SAS Label: Weight (kg)

English Text: Weight (kg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.7 to 248.2	Range of Values	8754	8754	
.	Missing	106	8860	

BMIWT - Weight Comment

Variable Name: BMIWT

SAS Label: Weight Comment

English Text: Weight Comment

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	30	30	
3	Clothing	296	326	
4	Medical appliance	19	345	
.	Missing	8515	8860	

BMXRECUM - Recumbent Length (cm)

Variable Name: BMXRECUM

SAS Label: Recumbent Length (cm)

English Text: Recumbent Length (cm)

Target: Both males and females 0 MONTHS - 47 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
48.5 to 118.8	Range of Values	454	454	
.	Missing	8406	8860	

BMIRECUM - Recumbent Length Comment

Variable Name: BMIRECUM

SAS Label: Recumbent Length Comment

English Text: Recumbent Length Comment

Target: Both males and females 0 MONTHS - 47 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	18	18	
3	Not straight	0	18	
.	Missing	8842	8860	

BMXHEAD - Head Circumference (cm)

Variable Name: BMXHEAD
SAS Label: Head Circumference (cm)
English Text: Head Circumference (cm)
Target: Both males and females 0 MONTHS - 6 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
34.4 to 46.5	Range of Values	70	70	
.	Missing	8790	8860	

BMIHEAD - Head Circumference Comment

Variable Name: BMIHEAD

SAS Label: Head Circumference Comment

English Text: Head Circumference Comment

Target: Both males and females 0 MONTHS - 6 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	0	0	
.	Missing	8860	8860	

BMXHT - Standing Height (cm)

Variable Name: BMXHT

SAS Label: Standing Height (cm)

English Text: Standing Height (cm)

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
79.1 to 200.7	Range of Values	8499	8499	
.	Missing	361	8860	

BMIHT - Standing Height Comment

Variable Name: BMIHT

SAS Label: Standing Height Comment

English Text: Standing Height Comment

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	46	46	
3	Not straight	88	134	
.	Missing	8726	8860	

BMXBMI - Body Mass Index (kg/m2)****Variable Name:** BMXBMI**SAS Label:** Body Mass Index (kg/m**2)**English Text:** Body Mass Index (kg/m**2)**Target:** Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11.1 to 74.8	Range of Values	8471	8471	
.	Missing	389	8860	

BMDBMIC - BMI Category - Children/Youth

Variable Name: BMDBMIC

SAS Label: BMI Category - Children/Youth

English Text: BMI Category - Children/Youth

Target: Both males and females 2 YEARS - 19 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Underweight	92	92	
2	Normal weight	1469	1561	
3	Overweight	372	1933	
4	Obese	559	2492	
.	Missing	6368	8860	

BMXLEG - Upper Leg Length (cm)

Variable Name: BMXLEG

SAS Label: Upper Leg Length (cm)

English Text: Upper Leg Length (cm)

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
24.9 to 51.6	Range of Values	7335	7335	
.	Missing	1525	8860	

BMILEG - Upper Leg Length Comment

Variable Name: BMILEG

SAS Label: Upper Leg Length Comment

English Text: Upper Leg Length Comment

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	396	396	
.	Missing	8464	8860	

BMXARML - Upper Arm Length (cm)

Variable Name: BMXARML
SAS Label: Upper Arm Length (cm)
English Text: Upper Arm Length (cm)
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10 to 49.2	Range of Values	8568	8568	
.	Missing	292	8860	

BMIARML - Upper Arm Length Comment

Variable Name: BMIARML

SAS Label: Upper Arm Length Comment

English Text: Upper Arm Length Comment

Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	200	200	
.	Missing	8660	8860	

BMXARMC - Arm Circumference (cm)

Variable Name: BMXARMC
SAS Label: Arm Circumference (cm)
English Text: Arm Circumference (cm)
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12 to 63.3	Range of Values	8562	8562	
.	Missing	298	8860	

BMIARMC - Arm Circumference Comment

Variable Name: BMIARMC
SAS Label: Arm Circumference Comment
English Text: Arm Circumference Comment
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	205	205	
.	Missing	8655	8860	

BMXWAIST - Waist Circumference (cm)

Variable Name: BMXWAIST
SAS Label: Waist Circumference (cm)
English Text: Waist Circumference (cm)
Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
39.8 to 187	Range of Values	8190	8190	
.	Missing	670	8860	

BMIWAIST - Waist Circumference Comment

Variable Name: BMIWAIST
SAS Label: Waist Circumference Comment
English Text: Waist Circumference Comment
Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	347	347	
.	Missing	8513	8860	

BMXHIP - Hip Circumference (cm)**Variable Name:** BMXHIP**SAS Label:** Hip Circumference (cm)**English Text:** Hip Circumference (cm)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
69.9 to 187.1	Range of Values	6776	6776	
.	Missing	2084	8860	

BMIHIP - Hip Circumference Comment

Variable Name: BMIHIP

SAS Label: Hip Circumference Comment

English Text: Hip Circumference Comment

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	361	361	
.	Missing	8499	8860	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Blood Pressure - Oscillometric Measurements (BPXO_L)

Data File: BPXO_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

This section provides data for three consecutive blood pressure (BP) and pulse measurements using an oscillometric device.

Eligible Sample

Participants aged 8 years and older were eligible. Participants with any of the following on both arms were excluded from the exam: rashes, gauze dressings, casts, edema, paralysis, tubes, open sores or wounds, withered arms, and A-V shunts. Also, women who have had an axillary nodal biopsy or resection, or a unilateral radical mastectomy, do not have their blood pressure measured in the affected arm.

Protocol and Procedure

All BP measurements are taken in the mobile examination center (MEC). Prior to obtaining BP measurements, upper arm circumference is measured, which is done to guide selection of cuff size.

After resting quietly in a seated position for 5 minutes, three consecutive BP measurements (systolic and diastolic) were taken 60 seconds apart using a digital upper-arm electronic blood pressure measurement device, Omron HEM-907XL. This device has been previously validated by both the Association for the Advancement of Medical Instrumentation (AAMI) and the International Protocol of the European Society of Hypertension for taking BP measurements in populations aged 13 years and older (White et. al., 2002; Ombani et. al., 2007; Ostchega et. al., 2010).

BP measurements were taken in the right arm unless specific conditions prohibited the use of the right arm, or if participants reported any reason that the BP measurements should not be taken in the right arm.

Details on the protocol for obtaining upper arm circumference and BP are described in the NHANES [August 2021–August 2023](#) procedure manuals.

Quality Assurance & Quality Control

The BP examiners were certified for BP measurement through a training program and followed a strict standardized BP protocol. In addition, examiners were observed to assure that BP readings were obtained following the standardized protocol. Health technician performance was also monitored using data reviews.

Data Processing and Editing

The following are some specifications used in capturing the BP data:

- Systolic BP cannot be greater than 300 mmHg;
- Systolic blood pressure must be greater than diastolic BP;
- If there is no systolic BP, there can be no diastolic BP. However, there can be a systolic measurement without a diastolic measurement.

Analytic Notes

After the 2017-2018 survey cycle, NHANES stopped collection of auscultatory BP measurements using a mercury sphygmomanometer. During the 2017-2018 survey cycle, blood pressure was collected using both auscultatory (with a mercury sphygmomanometer) and oscillometric (with the Omron HEM-907XL) methods. Additional details and results from the study, comparing auscultatory and oscillometric BP measurements, are available in an NCHS Vital and Health Statistics Series 2 Report at [NCHS website](#) (Ostchega et. al., 2021).

Exam sample weights should be used for analyses. Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

References

- Ombani S, Riva I, Giglio A, Caldara G, Groppelli A, Parati G. Validation of the Omron M5-I, R5-I and HEM-907 automated blood pressure monitors in elderly individuals according to the international protocol of the European Society of Hypertension. *Blood Press Monit* 12(4):233-42. 2007.
- Ostchega Y, Nwankwo T, Chiappa M, Wolz M, Graber J, Nguyen DT. Comparing blood pressure values obtained by two different protocols: National Health and Nutrition Examination Survey, 2017-2018. National Center for Health Statistics. *Vital Health Stat* 2(187). 2021.
- Ostchega Y, Nwankwo T, Sorlie PD, Wolz M, Zipf G. Assessing the validity of the Omron HEM-907XL oscillometric blood pressure measurement device in a national survey environment. *J Clin Hypertens (Greenwich)* 12(1):22-8. 2010
- White WB, Anwar YA. Evaluation of the overall efficacy of the Omron office digital blood pressure HEM-10. El Assaad MA, Topouchian JA, Darné BM, Asmar RG. Validation of the Omron HEM-907 device for blood pressure measurement. *Blood Press Monit* 7(4):237-41. 2002.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 8 YEARS - 150 YEARS

BPAOARM - Arm selected - oscillometric**Variable Name:** BPAOARM**SAS Label:** Arm selected - oscillometric**English Text:** Arm selected - oscillometric**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
L	Left	51	51	
R	Right	7603	7654	
< blank >	Missing	147	7801	

BPAOCSZ - Coded cuff size - oscillometric

Variable Name: BPAOCSZ

SAS Label: Coded cuff size - oscillometric

English Text: Mid arm circumference (cm) cuffing parameters - oscillometric

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2	17-21.9 (bladder size = 9.20 x 16.68 cm)	330	330	
3	22-31.9 (bladder size = 12.49 x 23.52 cm)	3404	3734	
4	32-41.9 (bladder size = 14.98 x 31.19 cm)	3442	7176	
5	42-50 (bladder size = 17.98 x 37.89 cm)	435	7611	
.	Missing	190	7801	

BPXOSY1 - Systolic - 1st oscillometric reading

Variable Name: BPXOSY1

SAS Label: Systolic - 1st oscillometric reading

English Text: Systolic - 1st oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
61 to 232	Range of Values	7517	7517	
.	Missing	284	7801	

BPXODI1 - Diastolic - 1st oscillometric reading

Variable Name: BPXODI1

SAS Label: Diastolic - 1st oscillometric reading

English Text: Diastolic - 1st oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
33 to 142	Range of Values	7517	7517	
.	Missing	284	7801	

BPXOSY2 - Systolic - 2nd oscillometric reading

Variable Name: BPXOSY2

SAS Label: Systolic - 2nd oscillometric reading

English Text: Systolic - 2nd oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
59 to 233	Range of Values	7505	7505	
.	Missing	296	7801	

BPXODI2 - Diastolic - 2nd oscillometric reading

Variable Name: BPXODI2

SAS Label: Diastolic - 2nd oscillometric reading

English Text: Diastolic - 2nd oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
32 to 139	Range of Values	7505	7505	
.	Missing	296	7801	

BPXOSY3 - Systolic - 3rd oscillometric reading

Variable Name: BPXOSY3

SAS Label: Systolic - 3rd oscillometric reading

English Text: Systolic - 3rd oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
50 to 232	Range of Values	7480	7480	
.	Missing	321	7801	

BPXODI3 - Diastolic - 3rd oscillometric reading

Variable Name: BPXODI3

SAS Label: Diastolic - 3rd oscillometric reading

English Text: Diastolic - 3rd oscillometric reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
24 to 136	Range of Values	7480	7480	
.	Missing	321	7801	

BPXOPLS1 - Pulse - 1st oscillometric reading

Variable Name: BPXOPLS1
SAS Label: Pulse - 1st oscillometric reading
English Text: Pulse - 1st oscillometric reading
Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
35 to 158	Range of Values	7517	7517	
.	Missing	284	7801	

BPXOPLS2 - Pulse - 2nd oscillometric reading

Variable Name: BPXOPLS2
SAS Label: Pulse - 2nd oscillometric reading
English Text: Pulse - 2nd oscillometric reading
Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
32 to 141	Range of Values	7505	7505	
.	Missing	296	7801	

BPXOPLS3 - Pulse - 3rd oscillometric reading

Variable Name: BPXOPLS3
SAS Label: Pulse - 3rd oscillometric reading
English Text: Pulse - 3rd oscillometric reading
Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
31 to 154	Range of Values	7480	7480	
.	Missing	321	7801	

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Liver Ultrasound Transient Elastography (LUX_L)

Data File: LUX_L.xpt

First Published: September 2024

Last Revised: NA

Component Description

Chronic liver disease and cirrhosis are significant contributors to morbidity and mortality in the U.S. population (Singh et al., 2013; Tapper and Parikh, 2018; Yoon, 2018). With the obesity epidemic, nonalcoholic fatty-liver disease is considered the most common cause of chronic liver disease in U.S. adults and children. Other important causes of chronic liver disease in the general population include alcoholic liver disease and chronic viral hepatitis infections (B or C).

The main goals of the NHANES liver ultrasound transient elastography (variable name prefix LUX) component are to provide objective measures for two important liver disease manifestations: liver fibrosis (scarring in the liver) and hepatic steatosis (fat in the liver). A healthy liver is usually soft and flexible, but a person with liver disease tends to have a liver that is stiff. Liver fibrosis was measured by FibroScan®, which uses ultrasound and the vibration controlled transient elastography (VCTE™) to derive liver stiffness. The device also simultaneously measures the ultrasound attenuation related to the presence of hepatic steatosis and records the controlled attenuation parameter (CAP™) as the indicator for the fatness in the liver. Elastography has been evaluated by others for its accuracy to assess liver steatosis and liver fibrosis (Tang et al., 2015, Castéra et al., 2005, Barr et al., 2015).

Eligible Sample

All participants 12 years and older in the NHANES August 2021-August 2023 sample were eligible. Participants were excluded if they (1) were unable to lie down on the exam table, (2) were pregnant (or unsure if pregnant) at the time of their exam, or a urine could not be obtained to test for pregnancy, (3) had an implanted electronic medical device, or (4) were wearing a bandage or had lesions on the right side of their abdomen by the ribs (where measurements would be taken).

Protocol and Procedure

A detailed description of the procedures was documented in the [August 2021 – August 2023 Liver Ultrasound Transient Elastography Procedures Manual](#) of this component. The elastography measurements were obtained in the NHANES Mobile Examination Center (MEC), using the FibroScan® model 502 V2 Touch equipped with a medium (M) or extra-large (XL) wand (probe).

With FibroScan®, a mechanical vibration of mild amplitude and low frequency (50Hz) is transmitted through the intercostal space using a vibrating tip contacting the skin. The vibration induces a shear wave that propagates through the liver. The displacements induced by the shear waves were tracked and measured using pulse echo ultrasound acquisition algorithms. The shear wave velocity is related directly to tissue stiffness; with harder tissues, there is faster shear wave propagation. Using the Young modulus, the velocity is converted into liver stiffness, and expressed in kilopascals. In systematic reviews comparing vibration controlled transient elastography (VCTE™) to biopsy (as a gold standard) for the detection of severe liver fibrosis, the mean area under the receiver operating characteristic (ROC) curve

was 0.89 (95% CI, 0.88-0.91) (Tsochatzis et al., 2011, Friedrich-Rust et al., 2008) and the overall sensitivity and specificity were 82% (95% CI, 78-86%) and 86% (95% CI 0.80-0.91). In addition to the high accuracy, meta-analyses have demonstrated FibroScan® results carry significant prognostic value (Singh et al., 2013). Transient elastography has been FDA approved as a test for the evaluation of liver fibrosis.

The FibroScan® machine has also incorporated a novel physical parameter (controlled attenuation parameter or CAP™), which measures the ultrasound attenuation related to the presence of hepatic steatosis. The CAP measurement is recorded simultaneously with the liver stiffness measurement. The accuracy of the CAP measurement for the detection of steatosis against biopsy has been reported in few studies; for steatosis $\geq 10\%$, the area under the ROC curve is 0.81, with a sensitivity and specificity of 76% and 79%; for steatosis $>34\%$ these values were 0.80, 79% and 71%, respectively (Myers et al., 2012, de Ledinghen V et al., 2016, Sasso et al., 2016).

The elastography exam was performed by NHANES examiners, including health technicians (HTs), radiology technicians (RTs), and clinicians (registered nurse), who were trained and certified by NHANES staff and the equipment manufacturer (Echosens™ North America). The exams were performed according to the manufacturer guidelines. To help maintain a standardized data quality, the machine conducts and displays several quality control (QC) measures during the test: 1) the median of all valid measurements performed during the examination; 2) interquartile range (IQR) which represents the interval around the median within which 50% of all valid measurements will fall; and 3) IQR/M: the ratio of the IQR to the median stiffness. All these QC indexes were recalculated after each new measurement. Examiners were trained to take 10 valid measurements with an IQR/M ratio less than 30%.

If the first 10 measurements taken had an IQR/M $\geq 30\%$, the examiner may choose to capture additional measurements until the IQR/M index was lower than 30%. The examiner also has the option to delete outlier measurements to lower the IQR/M from the list of valid measurements. It should be noted that examiners were only allowed to delete measurements from the beginning of the measurements. Once a measurement was chosen to be deleted, any measurements taken before the chosen one will be eliminated from the exam as well. This is to reduce bias, so examiners were unable to hand select which measurement(s) to delete.

The QC criteria of IQR/M $<30\%$ is to reduce variability and improve validity by taking measurements that result in few outliers. Multiple factors can affect the measurement and result in outliers, for example, a participant moves during the exam, or technical issues, such as the placement of the probe is not centered over the liver or perpendicular, or if structures such as lung or ribs appear while the measurement is captured. Participants with a lot of adipose tissue sometimes may make it difficult for examiners to capture 10 valid measurements. Because of the twelve-minute time limit and the challenge of locating the proper site to administer the exam on the body, examiners could end up retaining fewer than 10 valid measurements.

Quality Assurance & Quality Control

A detailed description of quality assurance and quality control measures considered for this component can be found in the [August 2021 – August 2023 Procedures Manual](#). Briefly, the NHANES examiners completed a 2-day training program with survey staff and an expert FibroScan® Technician (reference examiner). The training included an overview of the component and demonstrations conducted by the reference examiner with volunteer subjects. The reference examiner reviewed and demonstrated the proper technique of the FibroScan® examination. Supervised practice exercises followed, conducted with several volunteer subjects. The reference examiner would certify the examiner after observing 3 satisfactory exams.

NHANES staff members and an external university-based medical epidemiologist with expertise in chronic liver disease monitored MEC staff performance in the field through periodic visits and direct observations. Examiner performance was also monitored using data reviews for

each technician or clinician compared to all others and annual reference examiner refresher training.

Multiple times per year NHANES staff would select a sample of the original FibroScan® PDF files obtained by the examiners in the MEC for re-review. The samples selected for review include ones from new and experienced examiners, and participants with: 1) extreme stiffness (E), CAP, or E-IQR values, 2) stiffness (E) or CAP values that seemed unusual for younger participants, or 3) inconsistent extreme stiffness (E) and CAP values in the same person (i.e., low stiffness (E) and high CAP or high stiffness (E) and low CAP).

Annual FibroScan® wand calibration was performed by the manufacturer and software updates were performed according to manufacturer recommendations. In addition, NHANES used four shear wave liver fibrosis phantoms (CIRS Model 039) for determining variances within and between FibroScan® machines and probes over time.

Data Processing and Editing

Information obtained by staff in the MEC regarding pregnancy status/test results, fasting times, possible exam exclusions, and other comments were recorded in the NHANES database during the participant's MEC visit. All measures recorded by the FibroScan® machines were directly transferred via the Integrated Survey Information System to the NHANES database system immediately after each exam. Examiners were instructed to visually verify that the values transferred correctly.

Computerized data quality control procedures were performed to check for completeness and data validity and to identify logical inconsistencies and extreme data values (e.g., fasting times lasting more than 40 hours) and rare deviations in the protocol (e.g., technical error for number of measures recorded or duplicate files due to exam restarts).

Prior to data release, NHANES staff reviewed extreme values and cross-checked with other available data for verification and reviewed free-text comments noted by MEC staff and made edits or corrections as appropriate.

The liver elastography exam status code (LUAXSTAT) was created and indicates the following for each participant:

- 1 = Complete exam (i.e., fasting time of at least 3 hours, 10 or more complete stiffness (E) measures, and a liver stiffness interquartile (IQRe) range / median E <30%).
- 2 = Partial exam (i.e., either a fasting time < 3 hours, < 10 complete stiffness (E) measures, or a liver stiffness interquartile (IQRe) range/median E 30% or higher).
- 3 = Ineligible participant (see eligibility criteria above).
- 4 = Not done (i.e., refusal, limited time during exam visit, other).

The number of measures attempted (LUANMTGP) and the number of measures recorded (LUANMVGTP), using the final wand, were categorized at high end to 20 to 29, and 30 or more.

FibroScan® measures were not edited, and there are no imputed values in this file.

Elastography results were not reported to participants if they had <10 complete stiffness (E) measures or liver stiffness interquartile (IQRe) range/median E ≥30%, or fasted <3 hours, as recommended by the elastography equipment manufacturer. An exception to these criteria was permitted if the participant had an E value below the referral criteria and had at least 10 complete stiffness measures, even though the 3-hour fasting time was not satisfied. In this data file, elastography results are included for all participants regardless of the number of complete stiffness measures, the IQRe value, or the length of fast.

Analytic Notes

As stated above no changes were made to the stiffness, controlled attenuation parameter, IQRe, or IQRc values obtained from the FibroScan® machine. Analysts should be aware that

some extreme values may be present. Extreme values may be to the result of difficulty obtaining the measures due to participant body habitus (especially those who are obese or who have narrow intercostal spaces) or may represent truly high values.

Sample weights: the NHANES examination sample weights should be used to analyze elastography data unless it is merged with a more restrictive data file, such as the morning fasting sample, then use the sample weight appropriate for that more selective group.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorials](#) for further details on the use of sample weights and other analytic issues. Both are available on the NHANES website.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

LUAXSTAT - Elastography exam status

Variable Name: LUAXSTAT
SAS Label: Elastography exam status
English Text: Elastography exam status.
Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Complete	6280	6280	LUAPNME
2	Partial	514	6794	
3	Ineligible	201	6995	LUARXIN
4	Not done	204	7199	LUARXND
.	Missing	0	7199	

LUARXNC - Reason for partial exam

Variable Name: LUARXNC

SAS Label: Reason for partial exam

English Text: Reason for partial exam.

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Fasting < 3hrs	165	165	LUAPNME
2	Unable to obtain 10 valid measures	138	303	LUAPNME
3	IQR/M >30%	211	514	LUAPNME
.	Missing	6685	7199	

LUARXND - Reason exam not done

Variable Name: LUARXND**SAS Label:** Reason exam not done**English Text:** Reason exam not done.**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Participant refusal	106	106	LUANMTGP
2	Limited time	40	146	LUANMTGP
3	Other (e.g. physical or technical limitations)	58	204	LUANMTGP
.	Missing	6995	7199	

LUARXIN - Reason ineligible

Variable Name: LUARXIN

SAS Label: Reason ineligible

English Text: Reason ineligible.

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Pregnant/ Unable to get urine to test for pregnancy	67	67	LUANMTGP
2	Other (e.g. insulin pump or other implantable electronic device)	134	201	LUANMTGP
.	Missing	6998	7199	

LUAPNME - Exam wand type

Variable Name: LUAPNME

SAS Label: Exam wand type

English Text: Exam wand type.

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
M	M	5158	5158	
XL	XL	1574	6732	
< blank >	Missing	467	7199	

LUANMVGP - Count:complete measures from final wand

Variable Name: LUANMVGP

SAS Label: Count:complete measures from final wand

English Text: Total number of complete measures retained (using final wand).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	0	32	32	
1	1	13	45	
2	2	6	51	
3	3	8	59	
4	4	7	66	
5	5	11	77	
6	6	4	81	
7	7	7	88	
8	8	4	92	
9	9	14	106	
10	10	4945	5051	
11	11	974	6025	
12	12	334	6359	
13	13	138	6497	
14	14	89	6586	
15	15	51	6637	
16	16	21	6658	
17	17	15	6673	
18	18	11	6684	
19	19	9	6693	
20	20 to 29	31	6724	
30	30 or more	8	6732	
.	Missing	467	7199	

LUANMTGP - Count:measures attempted with final wand

Variable Name: LUANMTGP

SAS Label: Count:measures attempted with final wand

English Text: Total number of measures attempted (using final wand).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	Not done	467	467	
1	1	1	468	
2	2	1	469	
3	3	3	472	
5	5	1	473	
6	6	1	474	
7	7	2	476	
8	8	2	478	
9	9	4	482	
10	10	2361	2843	
11	11	1186	4029	
12	12	576	4605	
13	13	445	5050	
14	14	326	5376	
15	15	247	5623	
16	16	163	5786	
17	17	144	5930	
18	18	131	6061	
19	19	116	6177	
20	20-29	538	6715	
30	30 or more	484	7199	
.	Missing	0	7199	

LUXSMED - Median stiffness (E), kilopascals (kPa)

Variable Name: LUXSMED

SAS Label: Median stiffness (E), kilopascals (kPa)

English Text: Median liver stiffness (E). This indicator is presented with one digit to the right of the decimal ratio (e.g. XX.X), and the units for this measure are kilopascals (kPa).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.5 to 75	Range of Values	6700	6700	
.	Missing	499	7199	

LUXSIQR - Stiffness E interquartile range (IQRe)

Variable Name: LUXSIQR

SAS Label: Stiffness E interquartile range (IQRe)

English Text: Stiffness (E) interquartile range (IQRe) of final stiffness measures. This indicator is presented with one digit to the right of the decimal (e.g., XX.X).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 72.8	Range of Values	6687	6687	
.	Missing	512	7199	

LUXSIQRM - Ratio: Stiffness IQRe / median E

Variable Name: LUXSIQRM

SAS Label: Ratio: Stiffness IQRe / median E

English Text: Ratio of the stiffness IQRe / median E stiffness value. This indicator is presented as a percent with one digit to the right of the decimal (e.g., XX.X%).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1720.7	Range of Values	6687	6687	
.	Missing	512	7199	

LUXCAPM - Median CAP, decibels per meter (dB/m)

Variable Name: LUXCAPM

SAS Label: Median CAP, decibels per meter (dB/m)

English Text: Median controlled attenuated parameter (CAP). This indicator is presented as a whole number (e.g., XXX), and the units for this measure are decibels per meter (dB/m).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
100 to 400	Range of Values	6699	6699	
.	Missing	500	7199	

LUXCPIQR - CAP interquartile range (IQRc)

Variable Name: LUXCPIQR

SAS Label: CAP interquartile range (IQRc)

English Text: Controlled attenuated parameter (CAP) interquartile range (IQRc) of final CAP measures. This indicator is presented as a whole number (e.g., XX).

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 208	Range of Values	6686	6686	
.	Missing	513	7199	